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15 October 2026

The Editors
Genome Medicine

Dear Editors,

Please find enclosed my manuscript, *Chemical reachability limits mutagen-based TKI-resistance screens*, submitted as a Research Article.

The paper audits what the mutagen chemistries used in current (2024–2026) preclinical practice can and cannot produce, against two chemistry-free saturation mutagenesis atlases (Feng 2022 KRAS G12C; Oh 2026 ALK). We write a screen's expected output as $P(v) \propto S_{\text{mutagen}}(v) \cdot \exp(\sigma(v))$ and find that chemistry enters as a reachability threshold rather than as a ranking weight. A cytosine-plus-adenine base-editor pair carries an exactly zero rate for four of six substitution classes, which removes eight of twelve KRAS G12C panel variants and 80–100% of each ALK drug's chemistry-free top-10 from any such library. Two further panel variants (R68L, V8M), both reported in an accompanying *in vitro* scan rather than in a patient cohort, need two nucleotide changes and are suppressed 51- to 169-fold at practical library depth. Past that threshold the class weighting does not pay its way: ENU and Mutazyme II, whose internal class ratios differ 13-fold, give near-identical ALK top-100 recovery (61–67%), and on no anchor does $\log S + \sigma$ beat σ alone. We report that null as a result, and we report what the anchors can resolve (n = 4–11, minimum detectable gain 9–50 pp) rather than reading a non-significant p as evidence of no effect. The deliverable is an enumerable, codon- and spectrum-level inventory of blind spots, auditable before a screen runs; the Discussion names the specific variants the forthcoming neladalkib and next-generation KRAS G12C screens are likely to miss.

I believe the manuscript is a fit for *Genome Medicine* as a translational-methods paper: it uses genomic data to improve the preclinical screens that feed into clinical resistance panels and treatment-sequencing decisions, and its predictions are actionable both for screen designers and for clinicians interpreting preclinical-screen outputs when a patient relapses.

The work is single-authored, original, not under consideration elsewhere, and declared without competing interests. A bioRχiv preprint will accompany submission.

Suggested reviewers: James S. Fraser (UCSF; senior author on Estevam 2024 MET DMS); Jesse D. Bloom (Fred Hutchinson Cancer Center; DMS methodology); Serena Nik-Zainal (University of Cambridge; mutational signatures); Jessica J. Lin (Massachusetts General Hospital; clinical ALK resistance, first author on Lin 2024).

Sincerely,

Michele Ronco